

Propositions:

Week 1: Proposition 2.1 The properties (A0),..., (A4) above imply that, for $x, y, z \in \mathbb{R}$, **(a)** $x+y=x+z$ if and only if $y=z$ **(b)** If $x+y=x$, then $y=0$ **(c)** If $x+y=0$, then $x=-y$ **(d)** $(-(-x))=x$.

Proposition 2.2 The properties (M0),..., (M5) imply that, for $x, y, z \in \mathbb{R}$, if $x = 0$, (a) $xy=xz$ if and only if $y=z$ (b) If $xy=x$, then $y=1$ (c) If $xy=1$, then $y=1/x$ (d) $1/(1/x) = x$.

Proposition 2.3 For $x, y \in \mathbb{R}$, (a) $0 \cdot x = 0$. (b) If $x \neq 0 \neq y$, then $xy \neq 0$. (c) $(-x)y = -(xy) = x(-y)$. (d) $(-x)(-y) = xy$.

Proposition 2.6 Let a be rational and b be irrational. (a) $a+b$ is irrational. (b) if $a \neq 0$ then ab is irrational

Proposition 2.7 Given $x > 0$ and $n \in \mathbb{N}$, there is a unique $y > 0$ such that $y^n = x$, and we write this number y as $y = x^{1/n}$

Proposition 2.8 Let $x > 0$ and $p, q \in \mathbb{Q}$. Then (a) $x^p x^q = x^{p+q}$ (b) $(x^{(p)^q}) = x^{pq}$ (c) $(xy)^p = x^p y^p$.

Week 2: Proposition 3.1 if $x > 0$ then $-x < 0$

Proposition 3.2 If $x \neq 0$, then $x^2 > 0$.

Proposition 3.3 If $x > 0$, then $1/x > 0$.

Proposition 3.4 If $x > 0$, then $u > v$ if and only if $xu > xv$.

Proposition 3.5 If $0 < u, v$, then $u^2 > v^2$ if and only if $u > v$.

Lemma 3.6 For $u, v \in \mathbb{R}$, $uv \leq \frac{(u^2+v^2)}{2}$.

Proposition 3.7 AM-GM Inequality. Let $n \in \mathbb{N}$ and $x_1, \dots, x_n \geq 0$. Then $(x_1 \cdot x_2 \dots x_n)^{1/n} \leq \frac{(x_1 + \dots + x_n)}{n}$

Proposition 3.8 Cauchy-Schwarz Inequality. Let $n \in \mathbb{N}$, $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R}$. Then $x_1 y_1 + \dots + x_n y_n \leq \sqrt{x_1^2 + \dots + x_n^2} \sqrt{y_1^2 + \dots + y_n^2}$